

Lecture 27 — Finite Sets

CS 173: Discrete Structures · Functions & Cardinality

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1 Finite Sets and Cardinality Notation

Definition 1.1 (Finite Set). We say a set X is **finite** if

$$(\exists n \in \mathbb{N}) (|X| = |n|).$$

Here $|A| = |B|$ means there exists a bijection $f: A \rightarrow B$.

Definition 1.2 (Cardinality of a Finite Set). If $|X| = |n|$ and $n \in \mathbb{N}$, then

$$|X| := n = \{0, 1, 2, \dots, n-1\}.$$

2 Injections and Left Inverses

2.1 Setup

Suppose $f: A \rightarrow B$ is an injection. We wish to show there exists a function $g: B \rightarrow A$ such that

$$g \circ f = \text{id}_A.$$

Such a g is called a **left inverse** for f .

Definition 2.1 (Identity Function). The **identity function** on a set A is the function $\text{id}_A: A \rightarrow A$ defined by

$$\text{id}_A(a) := a \quad \text{for every } a \in A.$$

Definition 2.2 (Left Inverse). Let $f: A \rightarrow B$. A function $g: B \rightarrow A$ is a **left inverse** for f if

$$g \circ f = \text{id}_A,$$

i.e. for every $a \in A$, $g(f(a)) = a$.

2.2 Composition with the Identity

Remark. For any functions $h: A \rightarrow A$ and $k: A \rightarrow A$, composing with the identity gives

$$h = \text{id}_A \circ h, \quad \text{id}_A \circ k = k.$$

In particular, id_A is both a left and right identity for composition of functions $A \rightarrow A$.

3 Monomorphisms, Epimorphisms, and Isomorphisms

The following terminology packages the left/right inverse characterisations into standard names.

Definition 3.1 (Monomorphism (Monic)). Let $f: A \rightarrow B$. We say f is a **monomorphism** (or f is **monic**) if and only if

$$(\exists g: B \rightarrow A) (g \circ f = \text{id}_A).$$

Equivalently, f has a left inverse.

Definition 3.2 (Epimorphism (Epic)). Let $f: A \rightarrow B$. We say f is an **epimorphism** (or f is **epic**) if and only if

$$(\exists g: B \rightarrow A) (f \circ g = \text{id}_B).$$

Equivalently, f has a right inverse.

Definition 3.3 (Isomorphism (of Sets)). A function f is an **isomorphism** (of sets) if and only if f is both monic and epic:

$$f \text{ is an isomorphism} \iff f \text{ is monic} \wedge f \text{ is epic}.$$

4 Main Theorems

Theorem 4.1 (Injection $\Rightarrow$ Left Inverse & Injective Retract). For any sets A, B and any $f: A \rightarrow B$: if f is injective, then there exists $g: B \rightarrow A$ such that

$$g \text{ is surjective} \quad \text{and} \quad g \circ f = \text{id}_A.$$

Corollary 4.2. $|A| \leq |B| \Rightarrow |B| \geq |A|$.

Theorem 4.3 (Surjection $\Rightarrow$ Right Inverse & Injective Section). For any sets A, B and any

$f: A \rightarrow B$: if f is surjective, then there exists $g: B \rightarrow A$ such that

$$g \text{ is injective} \quad \text{and} \quad f \circ g = \text{id}_B.$$

Corollary 4.4. $|A| \geq |B| \Rightarrow |B| \leq |A|$.

5 Trichotomy of Cardinality

Theorem 5.1 (Axiom 7 — Trichotomy of Cardinality). For any sets A and B ,

$$|A| \leq |B| \vee |A| \geq |B|.$$

Equivalently, exactly one of the following holds:

$$(|A| < |B|) \vee (|A| > |B|) \vee (|A| = |B|).$$

Remark. Trichotomy says that cardinalities are always comparable. Note that $|A| < |B|$ means $|A| \leq |B|$ and $\neg(|A| \geq |B|)$, i.e. there is an injection $A \rightarrow B$ but no injection $B \rightarrow A$. Similarly $|A| > |B|$ and $|A| = |B|$ cover the remaining cases. This axiom is equivalent to the Axiom of Choice and is not provable from ZF alone.

6 Summary

1. A set X is **finite** iff $(\exists n \in \mathbb{N})(|X| = |n|)$, where $n = \{0, 1, \dots, n-1\}$.
2. $|A| = |B|$ iff there exists a bijection $A \rightarrow B$.
3. The **identity function** $\text{id}_A(a) = a$ is left and right neutral for composition.
4. A **left inverse** of $f: A \rightarrow B$ is $g: B \rightarrow A$ with $g \circ f = \text{id}_A$.
5. **Mono/Epi/Iso**: f is monic $\iff f$ has a left inverse; f is epic $\iff f$ has a right inverse; f is an isomorphism $\iff f$ is monic and epic.
6. Injection $\Rightarrow \exists$ surjective left inverse; corollary: $|A| \leq |B| \Rightarrow |B| \geq |A|$.
7. Surjection $\Rightarrow \exists$ injective right inverse; corollary: $|A| \geq |B| \Rightarrow |B| \leq |A|$.
8. **Trichotomy** (Axiom 7): for any A, B , $|A| \leq |B| \vee |A| \geq |B|$; equivalently, exactly one of $|A| < |B|$, $|A| > |B|$, $|A| = |B|$ holds.

Remark (Looking Ahead). With trichotomy in hand, the Cantor–Bernstein theorem ($|A| \leq |B|$ and $|B| \leq |A|$ implies $|A| = |B|$) will give us a robust framework for comparing the sizes of infinite sets.